

Lab 6:

From the following link extract the weather data:

<https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all>

- a) Create a MapReduce program to find average temperature for each year from NCDC data set.
- b) Find the mean max temperature for every month.

```
// AverageDriver.java
package temp;
public class AverageDriver {
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Average Calculation");
        job.setJarByClass(AverageDriver.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        job.setMapperClass(AverageMapper.class);
        job.setReducerClass(AverageReducer.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

// AverageMapper.java
public class AverageMapper extends Mapper<LongWritable,Text,Text,IntWritable> {
    public static final int MISSING = 9999;
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        String year = line.substring(15, 19);
        int temperature = (line.charAt(87) == '+')
            ? Integer.parseInt(line.substring(88, 92))
            : Integer.parseInt(line.substring(87, 92));
        String quality = line.substring(92, 93);
        if (temperature != MISSING && quality.matches("[01459]"))
            context.write(new Text(year), new IntWritable(temperature));
    }
}

// AverageReducer.java
public class AverageReducer extends Reducer<Text,IntWritable,Text,IntWritable> {
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {
        int sumTemp = 0, count = 0;
        for (IntWritable v : values) { sumTemp += v.get(); count++; }
        if (count > 0)
            context.write(key, new IntWritable(sumTemp / count));
    }
}

// MeanMaxMapper.java (Part b)
public class MeanMaxMapper extends Mapper<LongWritable,Text,Text,IntWritable> {
    public static final int MISSING = 9999;
```

```

public void map(LongWritable key, Text value, Context context)
    throws IOException, InterruptedException {
    String line = value.toString();
    String month = line.substring(19, 21);
    int temperature = (line.charAt(87) == '+')
        ? Integer.parseInt(line.substring(88, 92))
        : Integer.parseInt(line.substring(87, 92));
    String quality = line.substring(92, 93);
    if (temperature != MISSING && quality.matches("[01459]"))
        context.write(new Text(month), new IntWritable(temperature));
}
}

// MeanMaxReducer.java (Part b)
public class MeanMaxReducer extends Reducer<Text,IntWritable,Text,Text> {
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {
        int sumTemp = 0, count = 0, maxTemp = Integer.MIN_VALUE;
        for (IntWritable v : values) {
            int temp = v.get();
            sumTemp += temp; count++;
            if (temp > maxTemp) maxTemp = temp;
        }
        if (count > 0)
            context.write(key, new Text("mean="+sumTemp/count+" max="+maxTemp));
    }
}

// Execution
$ start-all.sh
$ hadoop fs -mkdir /weather
$ hadoop fs -copyFromLocal /home/hadoop/Desktop/1901.txt /weather/test.txt
$ hadoop jar AverageTemperature.jar AverageDriver /weather/test.txt /weather/output
$ hadoop fs -cat /weather/output/part-r-000000
$ hadoop jar Mean_Max_Temp.jar MMDriver /rgs/avtemp.txt /out8
$ hdfs dfs -cat /out8/*

2026-05-03 14:59:23,230 INFO mapreduce.Job: Running job: job_local91822813_0001
2026-05-03 14:59:24,581 INFO mapreduce.Job: Counters: 36
    Map input records=6565
    Map output records=6564
    Reduce output records=1

// hadoop fs -cat /weather/output/part-r-000000:
1901    46

// hdfs dfs -cat /out8/*:
01      4
02      0
03      7
04     44
05    100
06    168
07    219
08    198

```

09	141
10	100
11	19
12	3